Chapter test

28 Biotechnology

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which two enzymes are needed to produce recombinant plasmids?

   A. DNA polymerase and DNA ligase
   B. DNA polymerase and restriction enzyme
   C. restriction enzyme and DNA ligase
   D. restriction enzyme and RNA polymerase ☐

2. Bacteria use restriction enzymes to

   A. synthesize proteins.
   B. synthesize DNA.
   C. destroy foreign proteins.
   D. destroy foreign DNA. ☐

3. Which of the following is/are characteristic(s) of vectors used in recombinant DNA technology?

   (1) They are proteins.
   (2) They can replicate inside cells.
   (3) They can have foreign DNA inserted into them.

   A. (1) only
   B. (2) only
   C. (1) and (3) only
   D. (2) and (3) only ☐

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4. Which of the following statements about the production of human insulin by genetic engineering are correct?

(1) It involves the use of recombinant DNA technology.
(2) It has less risk of allergies than the traditional method.
(3) The cost of production is more expensive than the traditional method.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

5. Which of the following are applications of recombinant DNA technology?

(1) production of Golden Rice
(2) using bacteria to produce cheese
(3) gene therapy

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

6. Which of the following is an example of bioremediation?

A. using GM bacteria to produce human proteins
B. using bacteria to extract metals from ores
C. using bacteria to clean up oil spills
D. adding nitrogen-fixing bacteria to the soil before planting crops ☐

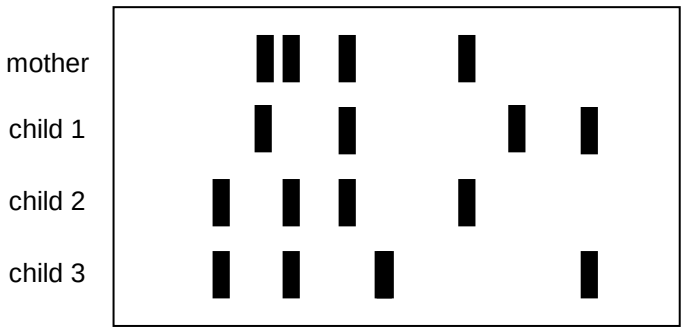
★ 7. Gel electrophoresis is useful in

(1) distinguishing between different alleles of a gene.
(2) identifying a virus.
(3) purifying specific DNA fragments.

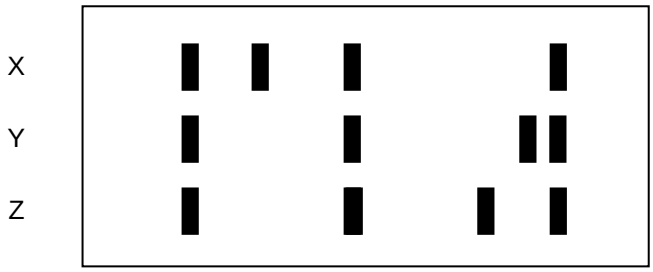
A. (1) only
B. (2) only
C. (1) and (3) only
D. (1), (2) and (3) ☐

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★ 8. The diagram below shows the DNA profiles of a mother and her three children.



Which of the following could be the DNA profile of the biological father of the three children?



A. X only
B. Y only
C. Z only
D. X and Z only ☐

★ 9. Which of the following correctly defines the human genome?

A. all of the human genes
B. all of the DNA in human cells
C. all of the DNA and RNA in human cells
D. none of the above ☐

10. Which of the following are the goals of the Human Genome Project?

(1) sequence human DNA
(2) map human genes
(3) allow transfer of genes from other species to humans

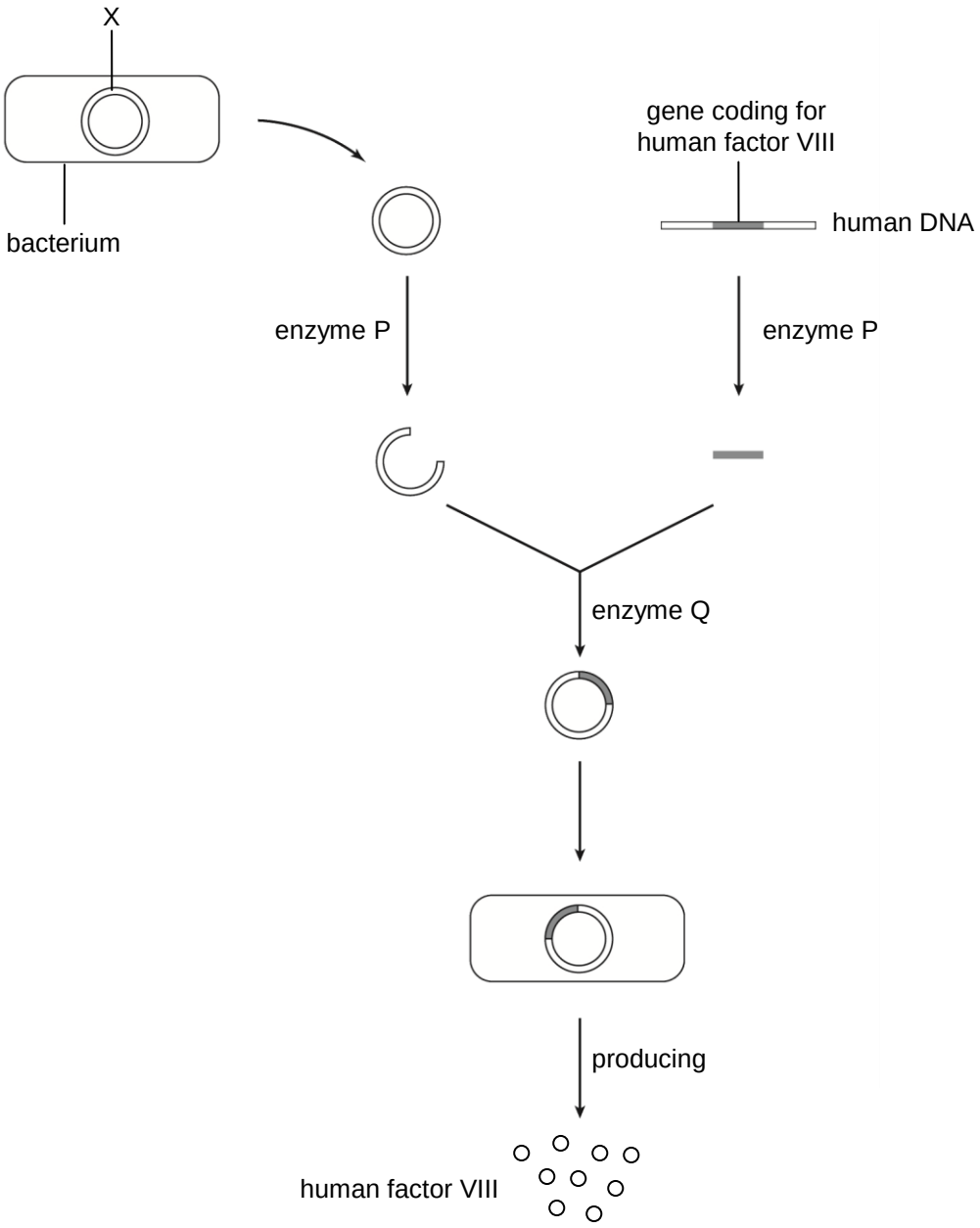
A. (1) and (2) only
B. (1) only (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

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Question	1	2	3	4	5	6	7	8	9	10
Related section	28.1	28.1	28.1	28.1	28.1	28.1	28.2	28.2	28.3	28.3

B. Problem solving (24 marks)

11. Human factor VIII is a protein necessary for the clotting of blood. The diagram below shows the major steps involved in producing human factor VIII using recombinant DNA technology.



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(a) (i) X is a small ring of DNA. Name X. (1 mark)

(ii) Describe the role of X in the process shown above. (3 marks)

(b) Identify enzymes P and Q. Give the function of each enzyme. (4 marks)

(c) Explain why human factor VIII can be produced in bacteria with the small ring of DNA introduced. (2 marks)

(Total: 10 marks)

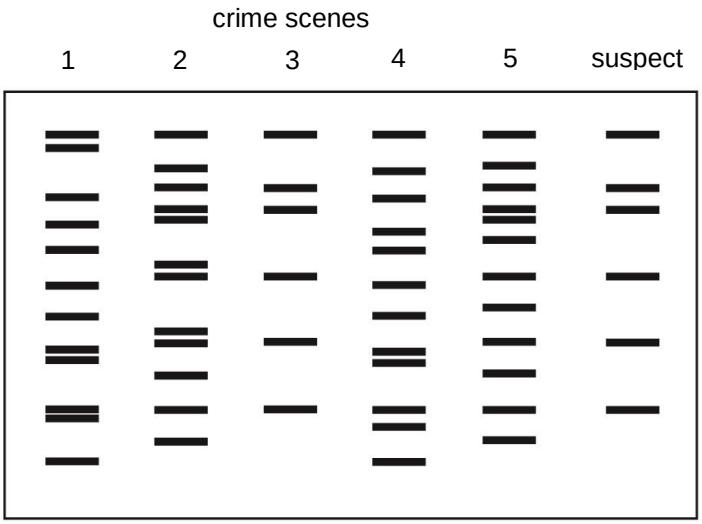
★ 12. DNA fingerprinting is used in forensic science.

(a) Give ***two*** examples of samples that can be collected from a crime scene for DNA fingerprinting. Explain your answer. (3 marks)

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(b) Describe how a DNA fingerprint is produced using the sample collected from a crime scene. (4 marks)

(c) The diagram below shows the DNA fingerprints produced using the samples collected from five crime scenes and a suspect. The samples collected from some of the crime scenes may contain DNA of more than one person.



In which crime scene(s) was the suspect present? Explain your answer. (2 marks)

(d) Give another application of DNA fingerprinting. (1 mark)

(Total: 10 marks)

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13. As a result of the Human Genome Project, scientists have developed genetic tests for many inherited diseases.

(a) State ***two*** advantages of the ability to test for inherited diseases. (2 marks)

(b) State ***two*** disadvantages of the ability to test for inherited diseases. (2 marks)

(Total: 4 marks)

C. Exam challenge (6 marks)

★★ 14. Scientists treated a DNA sample with enzyme X and then carried out gel electrophoresis on the sample.

(a) Enzyme X cut the DNA into fragments. What type of enzyme is enzyme X? (1 mark)

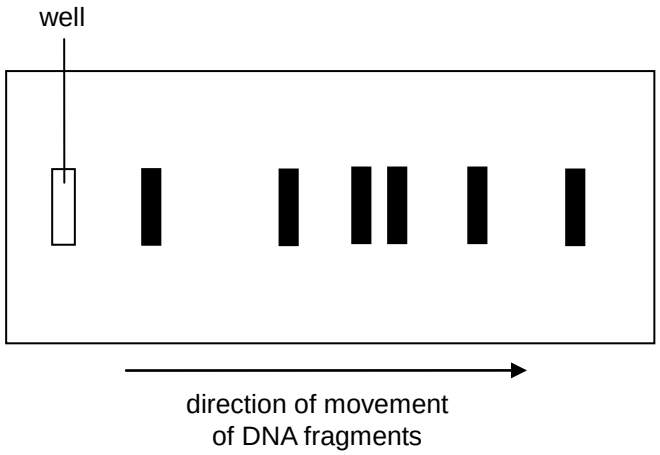
(b) The table below shows the length of the DNA fragments measured in the number of base pairs.

Fragment	Number of base pairs ($\times 10^3$)
1	6.98
2	8.58
3	16.07
4	3.59
5	8.03
6	11.30

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Explain why base pairs are a suitable way of measuring the length of a piece of DNA. (2 marks)

(c) The diagram below shows the results of gel electrophoresis.



(i) Circle the appropriate band on the diagram to show the position you would expect fragment 6 to have reached. (1 mark)

(ii) Explain how you arrived at your answer. (2 marks)

(Total: 6 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	D	3.	D	4.	A	5.	B
6.	C	7.	D	8.	C	9.	B	10.	A

(1 mark each)

B. Problem solving (24 marks)

11. (a) (i) Plasmid (1)

(ii) The plasmid acts as a vector. (1)

It carries the gene coding for human factor VIII (1)

and transfers it into bacteria. (1)

(b) P: restriction enzyme (1)

It is used to cut open the plasmid and the DNA fragment containing the human factor VIII gene. (1)

Q: DNA ligase (1)

It is used to catalyse the joining of the plasmid and the human factor VIII gene. (1)

(c) The gene in the plasmid is expressed in bacteria. (1)

Human factor VIII is produced by the process of protein synthesis using the machinery of the bacteria. (1)

(Total: 10 marks)

12. (a) Blood / saliva / semen / hairs (any ***two***) (1, 1)

Nucleated cells which contain DNA are present. (1)

(b) Extract DNA from the sample. (1)

Obtain DNA fragments containing different highly variable regions. (1)

Separate the DNA fragments according to their size using gel electrophoresis. (1)

The pattern of bands obtained (after staining) is a DNA fingerprint. (1)

(c) Crime scenes 2, 3 and 5 (1)

The DNA fingerprints produced using the samples collected from these crime scenes contain all the bands of the suspect's DNA fingerprints. (1)

(d) To establish family relationships in parentage tests / identify victim in disasters / screen for genetic diseases / authenticate Chinese medicines / trace of origin of products made from protected species / establish evolutionary relationships of organisms

(Accept other reasonable answers) (1)

(Total: 10 marks)

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13. (a) Early diagnosis can be carried out for prevention before symptoms appears. (1)
Prospective parents can know about the risks of having an affected child and make decisions concerning the affected pregnancy. (1)
(Accept other reasonable answers)

(b) It may cause psychological stress on the person if they are found to carry defective genes. (1)
It may lead to discrimination / abuse of knowledge, by employers / insurance companies. (1)
(Accept other reasonable answers)

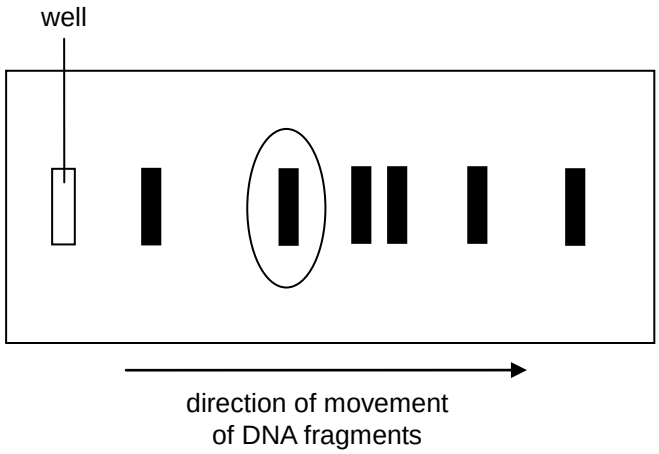
(Total: 4 marks)

C. Exam challenge (6 marks)

14. (a) Restriction enzyme (1)

(b) DNA is made of base pairs. (1)
Each base pair is of the same length / occupies same distance along the sugar-phosphate backbone. (1)

(c) (i)



(1)

(ii) Larger fragments travel more slowly than the smaller ones. (1)
Fragment 6 is the second largest fragment identified. (1)

(Total: 6 marks)

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